

LAGOSMART RETAIL SALES & PROFITABILITY ANALYSIS

End-to-End Data Analytics portfolio project

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Tools: Excel | Power query | MySQL| Power BI

Project Type: Synthetic Retail Analytics Project

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EXECUTIVE SUMMARY

Business Context

LagosMart Retail Sales & Profitability Analysis evaluates the performance of a Nigerian retail business across sales, product profitability, customer experience, and operational activities. The analysis was conducted using a synthetic, AI-generated dataset containing 5,008 transaction records initially. Following data cleaning and validation, the final analytical dataset contained 5,000 records.

Key Findings

The analysis identified significant differences in sales performance across regions and states, with the **Southwest emerging as the strongest-performing region and Northeast recording the lowest sales contribution**. Lagos also recorded the highest sales among the top 10 states analyzed.

Sales performance was stronger through the **online store channel**, while new customers generated more sales revenue and represented a larger proportion of the customer base than returning customers.

Product analysis revealed an important distinction between **sales performance and profitability**. Phones & Tablets generated the highest sales revenue but recorded the lowest average profit margin at approximately 7%, while Fashion recorded the highest average profit margin at approximately 34%. This indicates that sales volume alone is not sufficient for evaluating product performance.

Customer analysis showed differences in recorded ratings across product categories and regions, while the customer base remained weighted toward new customers. This presents an opportunity to strengthen customer retention and encourage repeat purchasing.

Operational analysis showed that the majority of orders were successfully completed or delivered. However, return rates varied by category, with Electronics recording the highest return rate at 16.0%, while Baby Products and Sport & Fitness recorded the lowest at 4.8%. Cash on Delivery also recorded a substantially higher cancellation rate than other major payment methods.

Business Priorities

Based on these findings, LagosMart should prioritize:

1. **Investigating regional and state-level sales differences**
2. **Managing profitability alongside sales growth**
3. **Reviewing high-return and low-profit products**
4. **Strengthening customer retention**
5. **Reducing Cash-on-Delivery cancellation risk**

6. **Improving operational exception management**
7. **Strengthening transaction-data quality**

1. DATA PREPARATION & METHODOLOGY

1.1 Project Overview

LagosMart is a Nigerian retail business operating through **online and physical sales channels**, offering a diverse range of products across categories including Electronics, Phones & Tablets, Fashion, Beauty & Personal Care, Home & Kitchen, Furniture, Groceries, Sports & Fitness, Baby Products, and Books & Stationery.

This project analyses LagosMart's retail transaction data to evaluate **sales performance, product profitability, customer behavior, and operational performance**. The objective is to identify key performance patterns, areas of concern, and opportunities for business improvement using data-driven analysis.

The dataset used in this project is a **synthetic, AI-generated dataset created for analytical and portfolio-development purposes**. Therefore, the findings demonstrate the application of data analytics techniques and should not be interpreted as the actual commercial performance of a real LagosMart business.

1.2 Business Objective

The analysis was designed to answer the following key business objectives:

- Evaluate overall sales and profitability performance.
- Identify high-performing and low-performing product categories and individual products.
- Examine sales performance across regions and states.
- Understand customer purchasing patterns and customer composition.
- Evaluate customer ratings across regions, categories, and customer types.
- Assess payment methods and their relationship with order cancellations.
- Analyze product return patterns.
- Evaluate delivery and order status performance.
- Identify areas where operational or commercial improvements could increase profitability and customer satisfaction.

The ultimate objective is to provide **data-driven recommendations that can support LagosMart's sales, product, customer, and operational decisions**.

1.3 Dataset Description

The analysis was conducted using a synthetic AI- generated **LagosMart retail dataset**, containing **5,008 transactional records** covering LagosMart's sales activities across Nigeria initially. Following data cleaning, duplicate-record review, validation, and data quality treatments, the final analytical dataset contained **5,000 records**.

The dataset contains information relating to:

- Transaction details
- Dates
- Product categories
- Product names
- Quantity
- Unit price
- Sales amount
- Discount
- Cost
- Profit
- Profit margin
- Customer type
- Customer ratings
- Region and state
- Store type
- Payment method
- Order status
- Delivery status
- Return information

The dataset was used as the single underlying source for the Excel, Power BI, and SQL analyses, with the cleaned version serving as the primary analytical dataset.

1.4 Data Cleaning & Preparation

Before conducting the analysis, the raw dataset was reviewed and prepared to improve data quality and ensure that the resulting analysis was reliable.

The following data-quality issues were identified during the preparation process:

Duplicate records

Transaction IDs were reviewed for duplication during the initial validation, resulting in **16 duplicate ID** occurrences. After further review, **8 records** were confirmed as true duplicate and removed, while the remaining repeated IDs containing different attribute values were retained for investigation.

Missing Values

- Missing and blank values were identified across relevant fields, including customer ratings, return reasons, and discount-related fields.
- Missing values were assessed based on the nature of each field before being handled appropriately.
- Blank values that could not be reliably inferred were retained or treated as missing rather than assigning unsupported values.

Data Type and Formatting

- Numeric fields such as quantity, unit price, sales, cost, profit, and discounts were standardized to appropriate numeric formats.
- Date fields were standardized to ensure accurate time-based analysis.
- Currency and percentage fields were formatted consistently.
- Inconsistent unit-price formatting, including currency symbols and separators, was standardized.

Data Consistency

- Calculated fields were reviewed against their expected business logic.
- Inconsistent records were identified and investigated before analysis.

Following the cleaning process, a **clean analytical dataset** was prepared for use in Power BI and SQL.

1.5 Data Validation

Data validation was performed to confirm that the cleaned data and calculated measures were logically consistent.

The following validation checks were conducted:

Sales Amount Validation

$$\text{Sales} = \text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount})$$

This was used to identify records where the recorded sales amount differed from the expected calculation.

Cost Validation

$$\text{Quantity} \times \text{Unit Cost}$$

Cost values were reviewed to ensure that they were logically consistent with the underlying transaction data.

Profit Validation

$$\text{Profit} = \text{Sales} - \text{Cost}$$

This check was used to verify the accuracy of recorded profit values.

Profit Margin Validation

$$\text{Profit Margin} = \text{Profit} \div \text{Sales}$$

Profit-margin calculations were validated and rounded appropriately to minimize discrepancies caused by decimal precision.

Operational Validation

Order, delivery, payment and return-related fields were also examined for inconsistencies and unusual values

These validation checks provided an additional layer of quality assurance and supported the reliability of the dataset for subsequent analysis.

1.6 Analytical Tools

Three main tools were used throughout the project.

Excel

Excel was used for:

- Initial data exploration
- Data cleaning
- Data validation
- Pivot-table analysis
- Exploratory analysis
- Preliminary visualizations

Power BI

Power BI was used to create the interactive analytical dashboard, covering:

- KPIs Cards
- Executive Overview
- Sales Performance
- Product Performance
- Customer Performance
- Operational Performance

Power BI was also used to create calculated measures, KPI cards, interactive visuals, and drill-down analysis and slicers.

MySQL

MySQL was used to perform SQL-based business analysis and validate findings and demonstrate the ability to derive business insights directly from the underlying data.

SQL queries were developed to answer business questions relating to sales, products, customers, profitability, returns, cancellations, and operational performance.

1.7 Calculated Metrics

The following key performance indicators were calculated to evaluate LagosMart's performance:

KPIS	Calculation
Total Sales	Sum of Sales Amount
Total Profit	Sum of Profit
Profit Margin	Total Profit ÷ Total Sales
Total Orders	Distinct Count of Transactional IDs
Average Customer Rating	Average of Customer Rating
Average Order Value	Total Sales ÷ Total Orders
Return Rate	Returned Orders ÷ Total Orders × 100
Cancellation Rate	Cancelled Orders ÷ Total Orders × 100

These metrics formed the basis of the **KPI cards** and supporting analyses presented in the Executive Overview of the Power BI dashboard.

1.8 Analytic Approach

The analysis followed a structured, business-question-driven approach rather than simply presenting descriptive statistics.

The cleaned dataset was first explored in Excel to understand the underlying data and identify key patterns and quality issues. Power BI was then used to develop an interactive dashboard that allowed performance to be examined across four core business areas:

Sales Performance

Analysis focused on sales by state, store type, payment method, customer type, category, and sales/profit trends.

Product Performance

Analysis examined profit by category, the most and least profitable products, top-selling products, and average profit margins.

Customer Performance

Analysis evaluated customer ratings, customer type distribution, and customer concentration across states.

Operational Performance

Analysis examined product returns, delivery status, payment-method cancellation rates, and delivery outcomes by payment method.

SQL analysis was subsequently used to query the cleaned dataset and answer business questions directly from the underlying data.

The findings from these analyses were interpreted using the following framework:

Business Question → Analysis → Key Finding → Business Insight → Recommendation

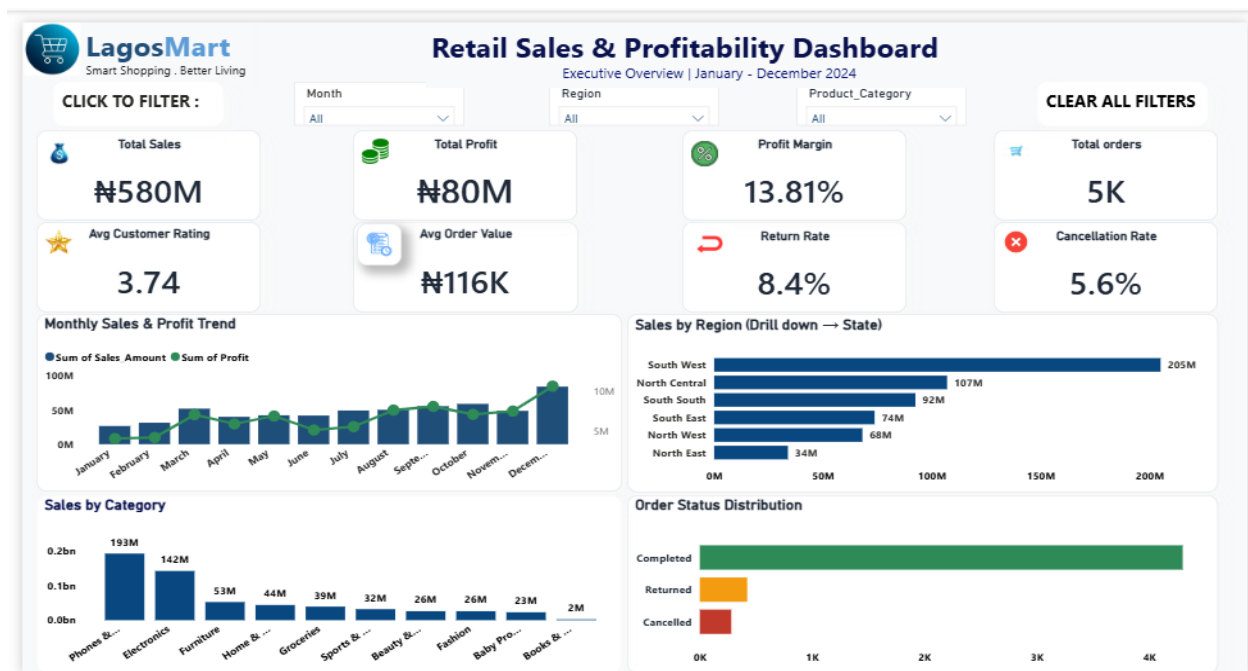
This approach ensured that the project moves beyond simply reporting numbers by identifying **patterns, potential business implications, and actionable recommendation.**

2. EXECUTIVE OVERVIEW

The Executive Overview provides a high-level view of LagosMart's overall business performance. It is designed to allow management or a recruiter reviewing the dashboard to understand the most important performance indicators and trends without having to examine every detailed analysis.

The page combines **eight key performance indicators with four summary visuals**, covering financial performance, sales trends, geographic performance, product categories, and order outcomes.

Figure 2.1: Lagosmart Power BI Executive Overview



2.2 Key Performance Indicators

The Executive Overview contains eight KPIs:

1. Total Sales

Measures the total revenue generated from recorded transactions.

2. Total Profit

Measures the overall profit generated after accounting for recorded costs.

3. Profit Margin

Measures the proportion of sales retained as profit.

Formula:

$$\text{Profit Margin} = \text{Total Profit} \div \text{Total Sales}$$

4. Total Orders

Formula:

Distinct count of Transaction ID

Measures the number of unique recorded transactions.

5. Average Customer Rating

Provides an overall indication of customer satisfaction based on recorded ratings.

6. Average Order Value

Formula:

$$\text{Average Order Value} = \text{Total Sales} \div \text{Total Orders}$$

This indicates the average monetary value of a transaction.

7. Return Rate

Formula:

$$\text{Return Rate} = \text{Returned Orders} \div \text{Total Orders} \times 100$$

This provides an indication of the proportion of orders associated with returns.

8. Cancellation Rate

Formula:

$$\text{Cancellation Rate} = \text{Cancelled Orders} \div \text{Total Orders} \times 100$$

This measures the proportion of orders that were cancelled.

Together, these KPIs provide a concise view of **revenue, profitability, transaction volume, customer experience, and operational performance.**

2.3 Executive Overview Visuals

The dashboard contains four summary visuals.

Monthly Sales & Profit Trend

Purpose:

To examine how sales and profit changed over time.

The combined sales-and-profit visual makes it possible to compare revenue generation with profitability rather than evaluating sales alone.

From the completed analysis:

- **December recorded the highest sales and profit.**
- **January recorded the lowest sales and profit.**

This indicates variation in both sales activity and profitability across the period analyzed.

Sales by Category

Purpose:

To compare the contribution of different product categories to overall sales. This enables management to quickly identify categories that contribute strongly to revenue and categories with comparatively lower sales contribution.

The detailed product analysis later examines profitability separately, because strong sales performance does not necessarily mean strong profitability.

Sales by Region with State Drill-Down

Purpose:

To understand geographical sales performance at two levels.

The visual initially provides a regional comparison and allow users to drill down from region to individual states.

This is particularly useful because it prevents the dashboard from displaying all 36 states simultaneously at the executive level, while still allowing detailed geographical analysis when required.

The detailed Sales Performance section subsequently examines the highest-performing states.

Order Status Distribution

Purpose:

To provide a high-level view of order outcomes.

The completed dashboard distinguishes the available order statuses:

- **Completed**
- **Return**
- **Cancelled**

This visual provides management with an immediate indication of how successfully recorded orders progressed through the order process.

2.4 Executive Overview

Business Purpose

The Executive Overview is intentionally designed as a **summary page rather than a detailed analytical page**.

It answers the high-level management question:

“How is LagosMart performing overall, and where should management investigate further?”

The eight KPIs provide the numerical performance snapshot, while the four visuals provide context around **sales composition, sales and profit trends, geographical performance and order outcomes**.

The detailed pages then allow the user to investigate the underlying drivers of these results across:

Sales → Products → Customers → Operations

3. SALES PERFORMANCE

3.1 Sales by Region

Business Question

Which region generated the highest sales revenue and which region underperformed?

See Executive Overview, Figure 2.1, for the regional sales visualization

Key Finding

The **South West** generated the highest sales revenue, while the **North East** recorded the lowest sales revenue.

Business Insight

South West emerged as LagosMart's strongest-performing region, while North East recorded the weakest revenue contribution, indicating a significant regional performance gap. This regional performance gap may reflect differences in customer demand, market reach, product availability or sales execution and warrants further investigation.

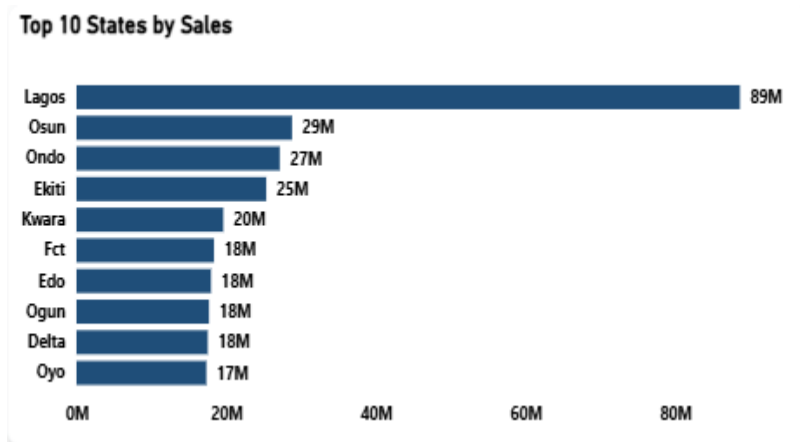
Recommendation

LagosMart should investigate the factors driving South West's strong performance and identify practices that can be adapted in other regions. A targeted review of the North East market should also be conducted to identify barriers to sales growth and develop region-specific improvement strategies.

3.2 Sales performance by State

Business Question

Which states recorded the highest sales revenue among the top-performing states?



Key Finding

Lagos recorded the highest sales revenue among the top 10 states analyzed, while Oyo recorded the lowest sales revenue within the top 10.

Business Insight

Sales performance varies substantially across states, with Lagos making the strongest contribution to revenue among the top-performing states. This indicates that revenue is concentrated in stronger markets and that differences in demand, market penetration, product mix, or sales execution may influence state-level performance.

Recommendation

Lagomart should investigate the factors contributing to Lagos's strong performance and assess which relevant practices can be adapted to other high-potential markets. Lower-performing states should be evaluated to identify barriers to sales growth before additional investment decisions are made.

3.3 Monthly Sales & Profit Trend

Business Question

How did LagosMart's sales revenue and profit change over time?

See Executive Overview, Figure 2.1, for the regional sales visualization

Key Findings

- **December recorded the highest sales revenue and highest profit** during the year.
- **January recorded the lowest sales revenue and lowest profit.**
- Overall, sales and profit generally moved in the same direction across the year, with stronger performance toward the later months.

Business Insight

LagosMart's sales and profitability strengthened toward the end of the year, with December recording the strongest performance and January the weakest. The concurrent movement of sales and profit indicates that periods of stronger sales performance were generally associated with stronger profitability generation during the year analyzed.

Recommendation

LagosMart should investigate the factors driving the strong December performance, such as customer demand, product mix, promotional activities and seasonal purchasing patterns, and assess which successful strategies can be replicated during weaker periods.

3.4 Sales by Product Category

Business Question

Which product categories generate the highest and lowest sales revenue for LagosMart?

See Executive Overview, Figure 2.1, for the regional sales visualization

Key Findings

Phones & Tablets generated the highest sales revenue among LagosMart's product categories, while Books & Stationery recorded the lowest sales revenue.

Business Insight

Sales revenue is concentrated in a number of stronger-performing product categories, with Phones & Tablets serving as the leading revenue contributor. The comparatively low sales contribution from Books & Stationery indicates weaker sales performance relative to other categories. The underlying drivers, such as demand, pricing, product assortment or availability, would require further investigation.

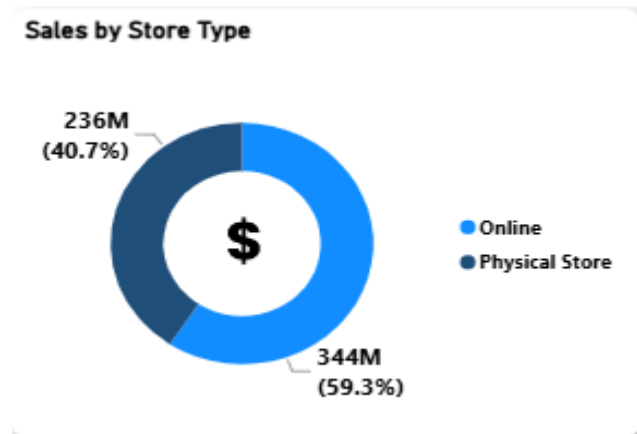
Recommendation

LagosMart should maintain adequate inventory and targeted marketing for high-performing categories such as Phones & Tablets while investigating the factors contributing to weak performance in Books & Stationery, including customer demand, pricing, product assortment and availability, before deciding whether to increase, reduce or reposition investment in the category.

3.5 Sales performance by Store Type

Business Question

Which store type contributes more to LagosMart's sales revenue: online or physical stores?



Key Findings

Online sales generated the majority of LagosMart's revenue, contributing approximately 59% of total sales, compared with approximately 41% from physical stores.

Business Insight

LagosMart's sales are more strongly driven by its online channel, indicating that digital purchasing is an important contributor to overall revenue. The lower contribution from physical stores suggests an opportunity to assess whether differences in customer reach, product availability and purchasing convenience are influencing channel performance.

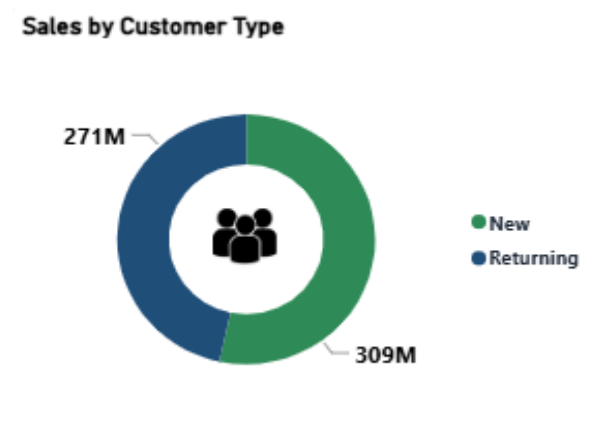
Recommendation

LagosMart should strengthen its online sales strategy by improving digital customer engagement, product visibility and purchasing convenience while evaluating opportunities to improve the performance of physical stores. Management should also investigate the factors driving the stronger online contribution before allocating additional resources between channels.

3.6 Sales by Customer Type

Business Question

How does sales performance differ between new and returning customers?



Key Finding

New customers generated higher sales revenue than returning customers, indicating that new customer acquisition contributed more to total sales than repeat purchases during the period analyzed.

Business Insight

LagosMart's sales performance is currently more heavily driven by new-customer acquisition than by revenue from returning customers. While this demonstrates an ability to attract new buyers, the comparatively lower contribution from returning customers may indicate an opportunity to strengthen customer retention and encourage repeat purchases.

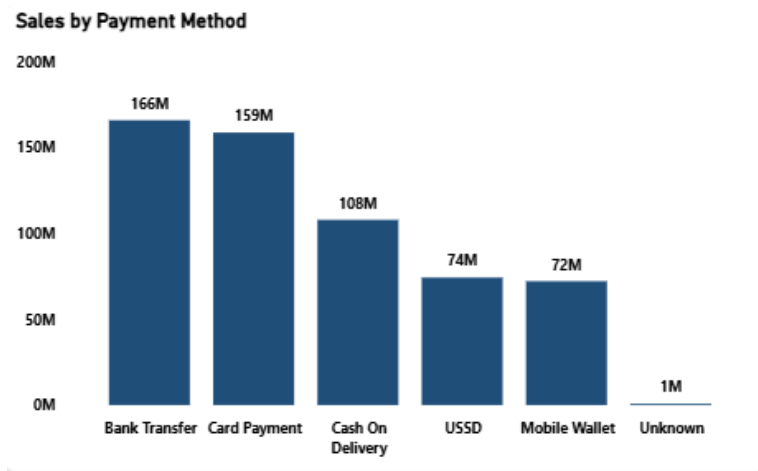
Recommendation

LagosMart should strengthen customer-retention through targeted loyalty initiatives, personalized promotions, post-purchase engagement designed to encourage first-time customers to make repeat purchases and increase customer lifetime value.

3.7 Sales by Payment Method

Business Question

Which payment methods generate the highest sales revenue for LagosMart?



Key Findings

Bank Transfer generated the highest sales revenue among the identified payment methods, while Mobile Wallet generated the lowest sales revenue among the identified named payment methods. As an additional “Unknown” category was also present in the dataset

Business Insight

Bank Transfer is the dominant payment method by sales revenue, indicating a strong customer preference for bank-based payments. In contrast, the relatively low revenue generated through Mobile Wallet suggests lower customer adoption of this payment option. The presence of an "Unknown" payment category also indicates that some payment-method records may require further data-quality investigation.

Recommendation

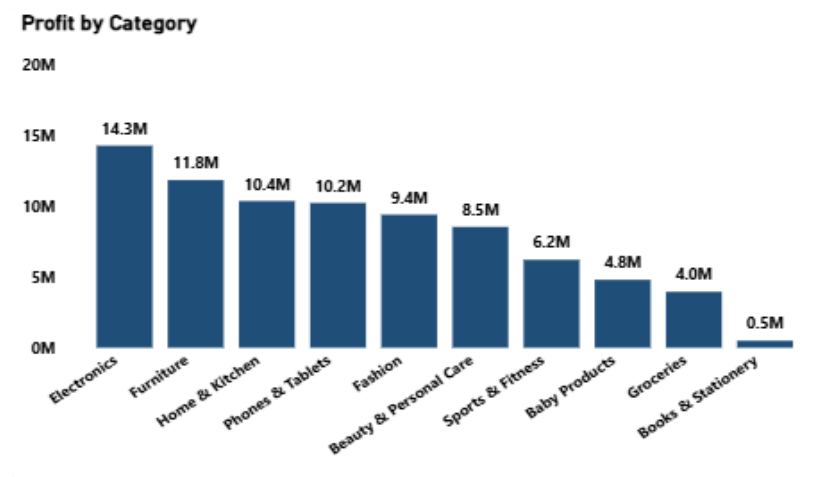
LagosMart should maintain and optimize Bank Transfer as a key payment option while investigating the low adoption of Mobile Wallet payments and review "Unknown" payment records to improve transaction-data accuracy. The business should also assess whether payment-processing issues, customer preferences, or checkout experience are contributing to the differences in payment-method usage.

4. PRODUCT PERFORMANCE

4.1 Profit by Product Category

Business Question

Which product categories generate the highest and lowest profit for LagosMart?



Key Findings

Electronics generated the highest profit among the product categories, while Books & Stationery recorded the lowest profit.

Business Insight

Profitability varies considerably across LagosMart's product categories. Electronics is the strongest contributor to overall profit, while Books & Stationery contributes the least, indicating that sales volume alone does not determine the financial value of a product category.

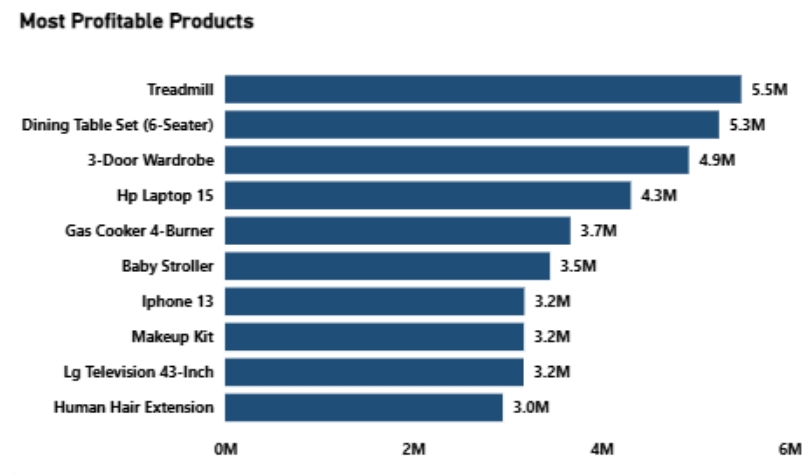
Recommendation

LagosMart should prioritize profitable categories such as Electronics while reviewing the pricing, costs, discounts and product mix of lower-profit categories such as Books & Stationery. Management should evaluate both revenue and profitability when making inventory and resource-allocation decisions rather than relying on sales volume alone.

4.2 Most Profitable Products

Business Question

Which individual products generate the highest profit for LagosMart?



Key Findings

The analysis identified the top 10 individual products by profit, with Treadmill recording the highest profit among the products analyzed.

Business Insight

Profitability varies considerably across individual products, with certain products contributing substantially more profit than others. These products may represent important opportunities for inventory prioritization, targeted marketing and revenue protection.

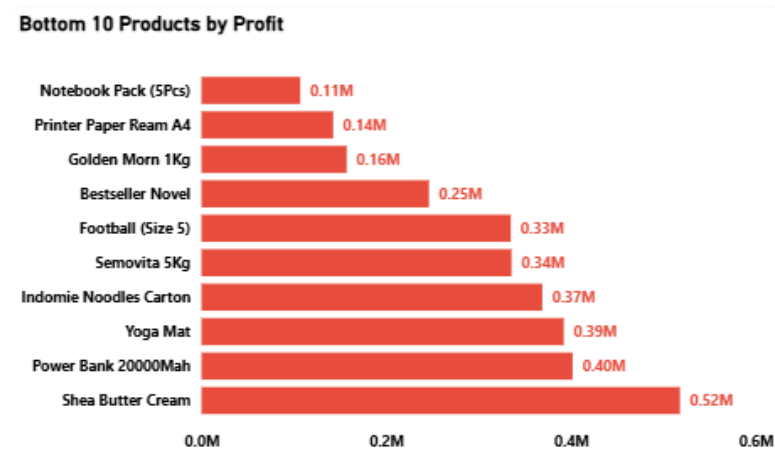
Recommendation

LagosMart should prioritize the availability and performance monitoring of its most profitable products while investigating the factors driving their profitability, including pricing, cost structure, demand and product mix.

4.3 Bottom 10 Products by Profit

Business Question

Which individual products contribute the least to LagosMart's profitability?



Key Findings

The bottom 10 products by profit generated substantially lower profit than the stronger-performing products, although they remained profitable during the period analyzed.

Business Insight

Profitability varies substantially across individual products, with the lowest-performing products contributing considerably less profit than the strongest products. Their lower profitability does not necessarily indicate that they should be discontinued, as the underlying drivers of their performance may include pricing, cost structure, discounts, or sales volume.

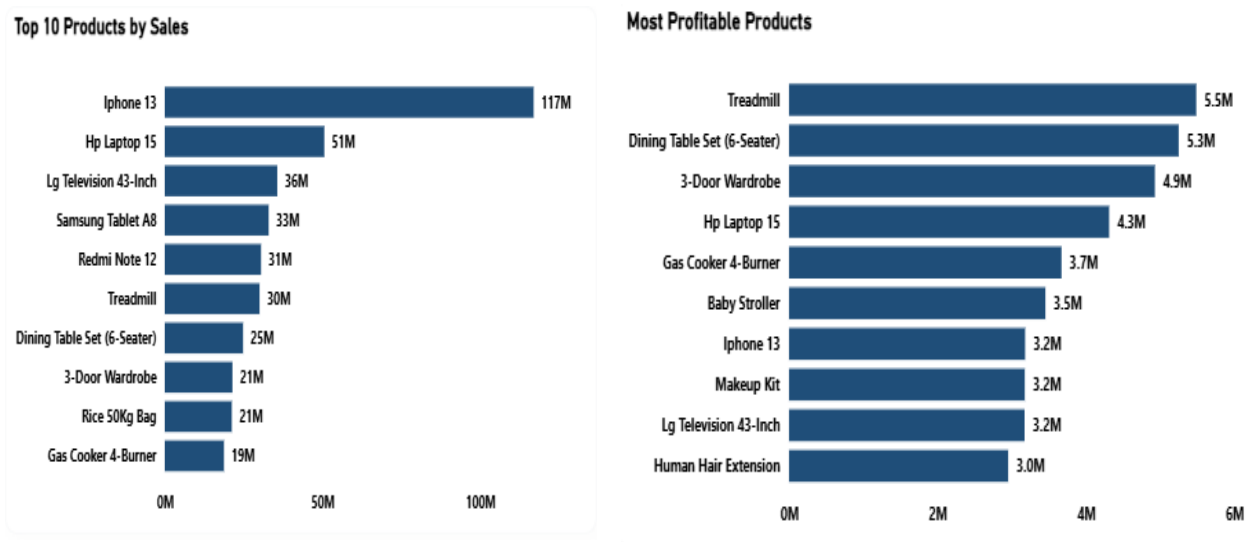
Recommendation

LagosMart should review the pricing, cost structure, discount levels, and sales volume of the lowest-profit products to identify opportunities for margin improvement. Management should assess the underlying cause of weak profitability before considering actions such as repricing, supplier renegotiation, product rationalization, or discontinuation.

4.4 Sales Revenue vs Product Profitability

Business Question

Do the products with the highest sales revenue also generate the highest profit?



Key Findings

The products generating the highest sales revenue were not necessarily the most profitable. For example, iPhone 13 ranked highly in sales but did not generate the highest profit, while Treadmill recorded the highest profit despite ranking sixth in sales.

Business Insight

The difference between sales and profit rankings demonstrates that sales volume alone is not a sufficient measure of product performance. Products with high sales may generate relatively lower profit because of factors such as pricing, product costs or discounts, while products with lower sales can generate stronger profit margins.

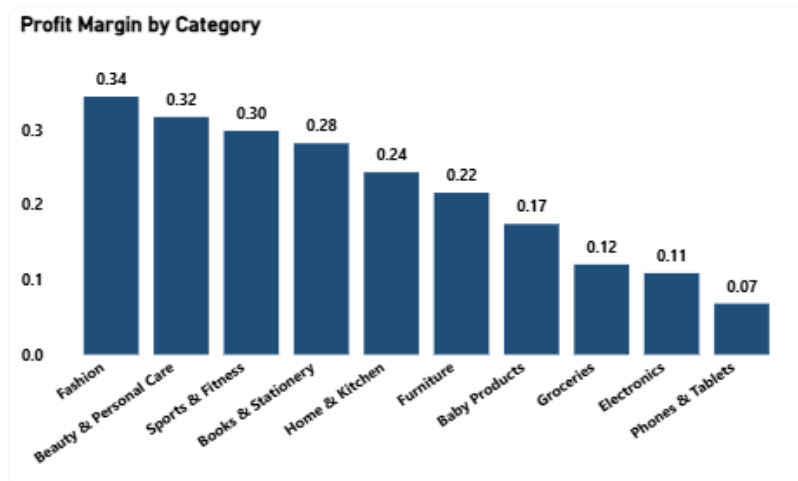
Recommendation

LagosMart should evaluate product performance using both sales revenue and profitability rather than sales volume alone. Management should prioritize products that provide a strong balance of revenue and profit while reviewing the cost, pricing and discount structures of high-sales but lower-profit products.

4.5 Average Profit Margin by Product Category

Business Question

Which product categories have the highest and lowest profit margins?



Key Finding

Fashion recorded the highest average profit margin at approximately 34%, while Phones & Tablets recorded the lowest at approximately 7%.

Business Insight

Profitability differs considerably across product categories. Although Phones & Tablets generated the highest sales revenue, the category recorded the lowest average profit margin, while Fashion achieved a substantially higher margin. This indicates that high sales revenue does not necessarily translate into high profitability per sale.

Recommendation

LagosMart should review the pricing, cost and discount structure of high-revenue but low-margin categories such as Phones & Tablets to identify opportunities to improve margins without significantly reducing demand. At the same time, the business should examine the factors behind Fashion's stronger margin and determine whether similar pricing or cost-management practices can be applied to other categories.

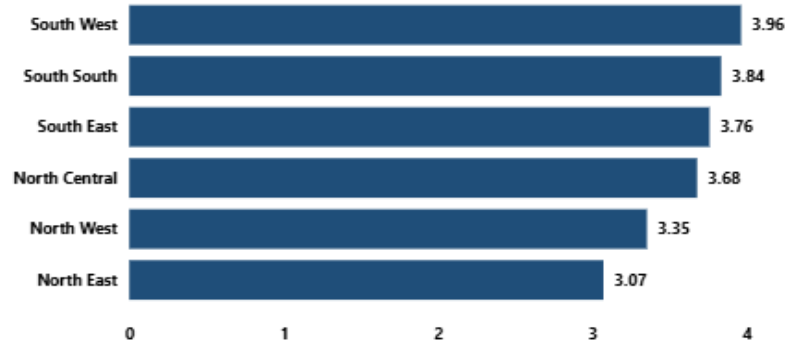
5. CUSTOMER INSIGHT

5.1 Customer Rating by Region

Business Question

How does customer satisfaction vary across LagosMart's regions?

Customer Rating by Region



Key Finding

South West recorded the highest average customer rating, while North East recorded the lowest, indicating differences in customer satisfaction across regions.

Business Insight

Customer satisfaction varies across LagosMart's regions, with South West demonstrating stronger customer experiences and North East recording comparatively low satisfaction. This regional variation suggests that factors affecting customer experience may differ across markets.

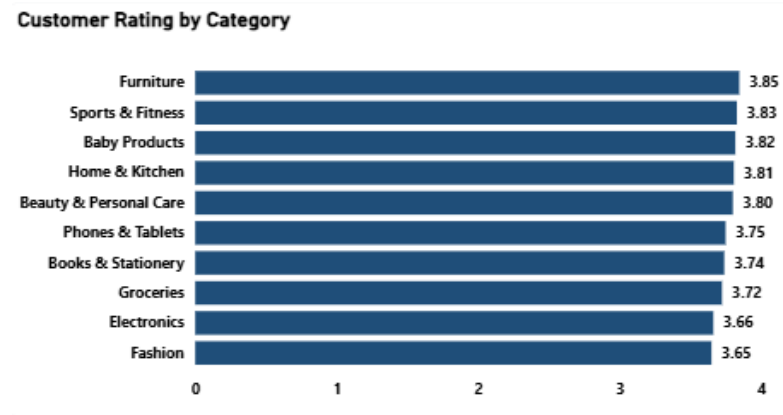
Recommendation

LagosMart should investigate whether differences in product availability, delivery experience, order fulfillment and service quality contribute to the regional variation in customer ratings.

5.2 Customer Rating by Category

Business Question

Which product categories receive the highest and lowest customer ratings?



Key Finding

Furniture recorded the highest average customer rating at 3.85, while Fashion recorded the lowest rating at 3.65.

Business Insight

Customer ratings were relatively consistent across product categories, with only modest differences between the highest-and lowest-rated categories. Furniture recorded the highest average rating, while the lowest- rated categories remained relatively close to the overall range.

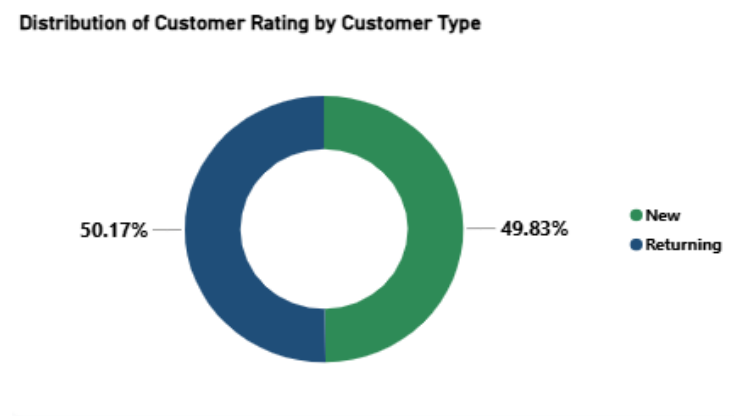
Recommendation

LagosMart should investigate the factors contributing to the strong ratings for Furniture and identify whether similar practices can be applied to lower-rated categories. The business should also review product quality, order fulfillment, delivery experience and customer feedback for Fashion to identify the main drivers of dissatisfaction.

5.3 Distribution of Customer Rating by Customer Type

Business Question

How are customer ratings distributed between new and returning customers?



Key Finding

New customers accounted for 50.17% of recorded customer ratings, while returning customers accounted for 49.83%, indicating a nearly even distribution of ratings between the two customer groups.

Business Insight

Customer feedback is relatively balanced between new and returning customers, suggesting that the available ratings provide perspectives from both customer groups rather than being dominated by one segment.

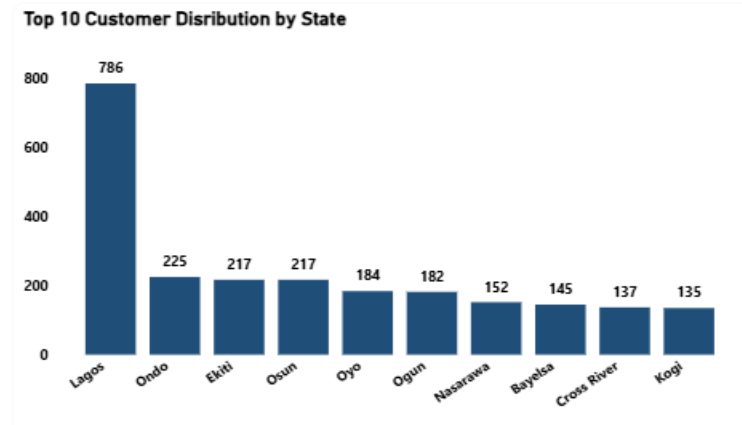
Recommendation

LagosMart should continue monitoring satisfaction across both customer groups and investigate whether the issues affecting new customers differ from those affecting returning customers. Segmenting customer feedback can help the business develop more targeted improvements to the customer experience.

5.4 Top 10 States by Customer Distribution

Business Question

Which states have the highest customer concentration?



Key Finding

Lagos had the highest customer concentration among the top 10 states, with 786 customers, substantially exceeding Kogi, which had 135 customers.

Business Insight

Customer concentration is uneven across states, with Lagos accounting for a substantially larger customer base than the other leading states. This indicates that customer reach is concentrated in a limited number of markets.

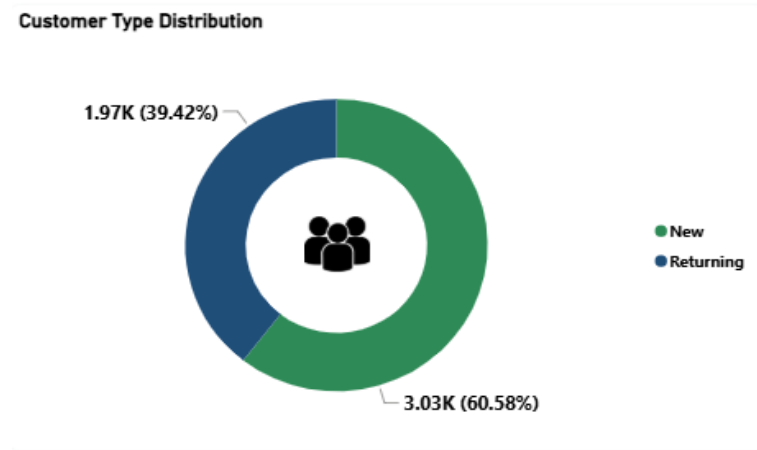
Recommendation

LagosMart should continue strengthening customer engagement in high-concentration markets while evaluating opportunities to expand customer reach in states with lower representation. Expansion decisions should be guided by market demand, sales potential and operational feasibility rather than customer count alone.

5.5 Customer Type Distribution

Business Question

How is LagosMart's customer base distributed between new and returning customers?



Key Finding

New customers accounted for approximately 60.58% of the customer base, while returning customers represented approximately 39.42%.

Business Insight

LagosMart's customer base is weighted toward new customers, indicating stronger customer acquisition than repeat-customer representation during the period analyzed. While the ability to attract new customers is positive, the lower proportion of returning customers highlights an opportunity to strengthen customer retention.

Recommendation

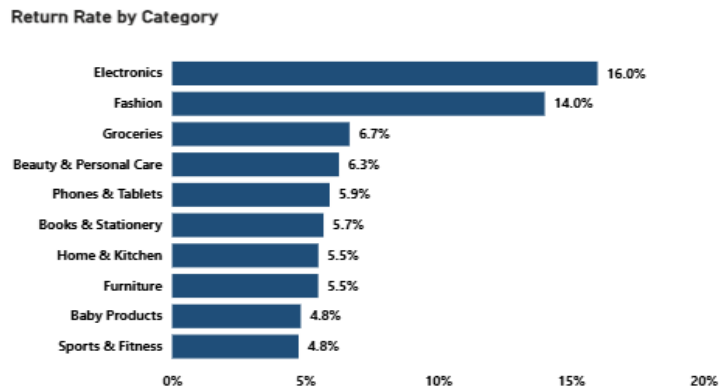
LagosMart should complement customer acquisition efforts with retention strategies such as loyalty programs, personalized offers and post-purchase engagement designed to convert first-time buyers into repeat customers.

6. OPERATIONAL PERFORMANCE

6.1 Return Rate by Category

Business Question

Which product categories have the highest and lowest return rates?



Key Finding

Electronics recorded the highest return rate at 16.0%, while Baby products and Sports & Fitness recorded the lowest at 4.8%.

Business Insight

Return rates vary substantially across product categories. Electronics had more than three times the return rate of Baby Product and Sports & Fitness. This indicates that returns are more concentrated in certain categories and may reflect differences in product quality, customer expectations, product descriptions or fulfillment.

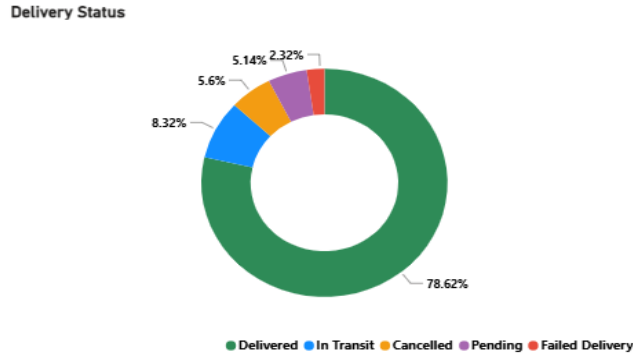
Recommendation

LagosMart should investigate the main reasons for returns in Electronics by reviewing return reasons, product quality, product descriptions and fulfillment processes. The business should also monitor return patterns across categories to identify recurring operational issues and reduce avoidable returns. Management should identify the practices associated with the lower return rate in Baby Product and Sports & Fitness and assess whether relevant practices can be applied to higher-return categories.

6.2 Delivery Status Distribution

Business Question

What is the distribution of delivery outcomes across LagosMart's orders?



Key Finding

Delivered orders accounted for 78.62% of total orders, while 8.32% remained in transit, 5.6% were cancelled, 5.14% were pending and 2.32% failed.

Business Insight

The majority of LagosMart's orders were successfully delivered, indicating generally strong order fulfillment. However, the remaining orders represent potential operational friction, particularly those that were cancelled, pending, in transit or failed.

Recommendation

LagosMart should monitor undelivered orders through delivery-status tracking and investigate the causes of cancellations, failed deliveries and prolonged pending or in-transit orders. Reducing these exceptions could improve fulfillment efficiency and customer experience.

6.3 Payment Method vs Cancellation

Business Question

Which payment methods are associated with the highest order cancellation rates?

Payment Method vs Cancellation Rate

Payment_Method	Total Orders	Cancelled Order	Cancellation Rate
Bank Transfer	1300	31	2.4%
Card Payment	1193	22	1.8%
Cash On Delivery	1146	175	15.3%
Mobile Wallet	625	18	2.9%
Unknown	12	1	8.3%
USSD	724	33	4.6%

Key Finding

Cash on Delivery recorded the highest cancellation rate at 15.3%, substantially higher than the other major payment methods. Card payments recorded the lowest cancellation rate at 1.8%.

Business Insight

Cash on Delivery was associated with a substantially higher cancellation rate than the other major payment methods, indicating that orders using this payment method may be more vulnerable to cancellation. The pattern warrants investigation into factors such as customer commitment, order confirmation and delivery-related issues.

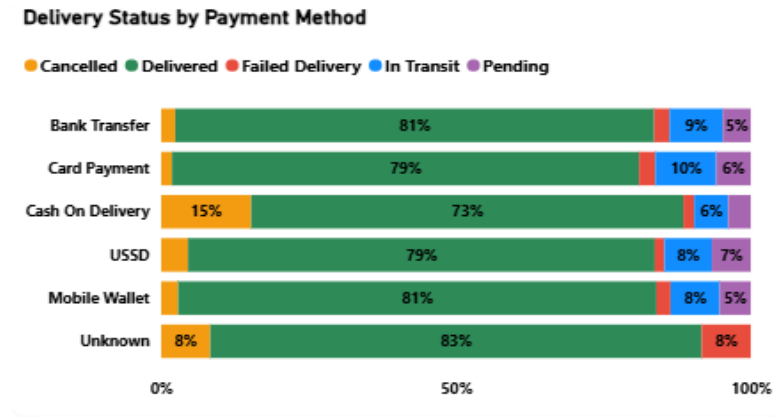
Recommendation

LagosMart should investigate the drivers of Cash-on-Delivery cancellations, including order confirmation, customer commitment and failed delivery attempts. The business should also strengthen order verification and consider appropriate incentives for reliable payment methods while retaining payment flexibility for customers.

6.4 Delivery Status by Payment Method

Business Question

How does order delivery performance vary across payment methods?



Key Finding

Delivery performance varied across payment methods, with Cash on Delivery recording the lowest delivered proportion at approximately 73% and the highest cancellation share at 15%. Bank Transfer and Mobile Wallet recorded stronger delivery performance, with approximately 81% of orders delivered.

Business Insight

Cash on Delivery was associated with weaker fulfillment outcomes, than the other major payment methods, driven primarily by its substantially higher cancellation rate. This suggests that the payment method may be associated with greater order commitment or fulfillment challenges.

Recommendation

LagosMart should investigate the Cash-on-Delivery order journey, particularly order confirmation, customer commitment and failed delivery attempts, to determine the causes of its higher cancellation rate. The business should also continue monitoring delivery performance across payment methods to identify opportunities for improving fulfillment efficiency.

6.5 Order Status Distribution

Business Question

What is the overall outcome of LagosMart's orders?

See Executive Overview, Figure 2.1, for the regional sales visualization

Key Finding

The majority of LagosMart orders were completed, with approximately 4.3K completed orders, while returned and cancelled orders represented substantially smaller portions of the total order volume.

Business Insight

LagosMart successfully completes the majority of its orders, indicating generally positive overall order fulfillment. However, the presence of cancelled and failed orders represents potential revenue leakage and operational inefficiency that should be monitored.

Recommendation

LagosMart successfully completed the majority of recorded orders, indicating generally positive order outcomes. However, returned and cancelled orders represent areas of potential revenue leakage and should be monitored.

7. KEY BUSINESS RECOMMENDATIONS

7.1 Strengthen Regional and State-Level Sales Performance

LagosMart should investigate the factors contributing to the stronger sales performance of the Southwest region and high-performing states such as Lagos. Relevant practices should be assessed for adaptation in other high-potential markets.

At the same time, lower-performing regions and states should be evaluated to identify barriers related to customer demand, market penetration, product availability, and sales execution before additional resources are allocated.

7.2 Prioritize Profitability Alongside Sales Growth

The analysis demonstrates that high sales volume does not necessarily translate into high profitability. Phones & Tablets generated the highest category sales but recorded the lowest average profit margin, while Fashion recorded a substantially higher margin.

LagosMart should therefore evaluate product performance using both **revenue and profitability**, with particular attention to pricing, product costs, discounts, and product mix.

7.3 Review Low-Profit and High-Return Products

Management should conduct product-level reviews of low-profit products and categories with relatively high return rates.

For products with weak profitability, the review should consider pricing, cost structure, discount levels, and sales volume. For high-return categories such as Electronics, management should examine return reasons, product quality, product descriptions, and fulfillment processes to identify opportunities to reduce avoidable returns.

7.4 Strengthen Customer Retention

New customers represented a larger proportion of the customer base and generated higher sales than returning customers.

While strong acquisition is positive, LagosMart should complement acquisition efforts with retention initiatives such as **loyalty programs, personalized offers, targeted promotions, and post-purchase engagement**.

The objective should be to convert more first-time buyers into repeat customers and increase customer lifetime value.

7.5 Address Cash-on-Delivery Cancellation Risk

Cash on Delivery recorded a substantially higher cancellation rate than the other major payment methods.

LagosMart should investigate the Cash-on-Delivery customer journey, particularly **order confirmation, customer commitment, and failed delivery attempts**. Strengthening order verification and monitoring cancellation patterns could help reduce avoidable cancellations while maintaining payment flexibility for customers.

7.6 Improve Operational Exception Management

Although most orders were successfully completed or delivered, cancelled, failed, pending, and in-transit orders represent areas of potential operational inefficiency.

LagosMart should establish regular monitoring of these exception categories and investigate recurring causes. Reducing avoidable delivery and order failures could improve fulfillment efficiency, customer experience, and revenue realization.

7.7 Strengthen Data Quality Monitoring

The presence of an **Unknown payment-method category** highlights the importance of maintaining accurate transaction records.

LagosMart should strengthen data-entry and transaction-record validation processes to reduce incomplete or unidentified payment records and improve the reliability of future business reporting.

CONCLUSION

The LagosMart analysis provides a multidimensional view of the business across **sales, profitability, customers, and operations**.

The analysis shows that sales performance is not evenly distributed across regions, states, customer groups, store types, and product categories. The Southwest region and Lagos demonstrated particularly strong sales performance, while online sales contributed a larger share of revenue than physical stores.

However, the analysis also demonstrates that **strong sales performance does not necessarily indicate strong profitability**. Phones & Tablets generated the highest category sales but recorded the lowest average profit margin, while Fashion achieved a substantially stronger margin. This highlights the importance of evaluating both revenue and profitability when making product and resource-allocation decisions.

Customer analysis showed a customer base weighted toward new customers, creating an opportunity for LagosMart to strengthen retention and encourage repeat purchases. Customer ratings also varied across regions and product categories, suggesting that customer experience should be monitored at a more granular level.

Operationally, the majority of orders were successfully completed or delivered. However, returns, cancellations, failed orders, and other delivery exceptions remain areas requiring attention. In particular, **Cash on Delivery was associated with a substantially higher cancellation rate**, making it a priority area for further investigation.

Overall, the analysis suggests that LagosMart has opportunities to improve performance by **protecting strong revenue-generating markets, improving product profitability, strengthening customer retention, reducing avoidable returns and cancellations, and improving operational and transaction-data quality**.

The project demonstrates how Excel, SQL, and Power BI can be combined to move from raw transactional data to structured business analysis and actionable recommendations.

Appendix

Appendix A – SQL Analysis Summary

SQL analysis was conducted using MYSQL to answer business questions relating to sales performance, product profitability, customer behavior, returns, cancellations and operational performance. The SQL analysis served as an additional validation and analytical layer over the cleaned LagosMart dataset.

Overall Business Performance

```
SUM(Sales_Amount) AS Total_Sales,  
  
SUM(Cost_Amount) AS Total_Cost,  
  
SUM(Profit) AS Total_Profit,  
  
ROUND(SUM(Profit) / NULLIF(SUM(Sales_Amount), 0) * 100, 2)  
  
AS Overall_Profit_Margin,  
  
COUNT(DISTINCT Transaction_ID) AS Total_Orders,  
  
ROUND(  
  
SUM(Sales_Amount) /  
  
NULLIF(COUNT(DISTINCT Transaction_ID), 0),  
  
2  
  
) AS Average_Order_Value  
  
FROM clean_lagosmart_data;
```

Business Purpose:

Identifies the overall sales, cost, and profit performance of LagosMart

Sales by Region

```
SELECT  
  
Region,  
  
SUM(Sales_Amount) AS Total_Sales
```

```
FROM clean_lagosmart_data
```

```
GROUP BY Region
```

```
ORDER by Total_SaleS DESC;
```

Business Purpose:

Identifies highest and lowest-performing regions.

Top Products by Profit

```
SELECT
```

```
SELECT
```

```
    Product_Name,
```

```
    SUM(Profit) AS Total_Profit
```

```
FROM clean_lagosmart_data
```

```
GROUP BY Product_Name
```

```
ORDER BY Total_Profit DESC
```

```
LIMIT 10;
```

Business Purpose:

Identifies the individual products generating the highest total profit.

Sales by Customer Type

```
SELECT
```

```
Customer_Type,
```

```
SUM(Sales_Amount) AS Total_Sales
```

```
FROM clean_lagosmart_data
```

```
GROUP BY Customer_Type
```

```
ORDER by Total_SaleS Desc;
```

Business Purpose:

Identified if New or Returning customers was associated with higher sales.

Cancellation Rate by Payment Method

```
SELECT

    Payment_Method,

    COUNT(CASE WHEN Order_Status = 'Cancelled' THEN 1 END) AS
Cancelled_Orders,

    COUNT(*) AS Total_Orders,

    ROUND(

        COUNT(CASE WHEN Order_Status = 'Cancelled' THEN 1 END)

        / COUNT(*) * 100,

        2

    ) AS Cancellation_Rate

FROM clean_lagosmart_data

GROUP BY Payment_Method

ORDER BY Cancellation_Rate DESC;
```

Business Purpose:

Identifies payment methods associated with higher cancellation rates.

Return Rate by Category

```
SELECT

    Product_Category,

    COUNT(CASE WHEN Order_Status = 'Returned' THEN 1 END) AS Returned_Orders,

    COUNT(*) AS Total_Orders,
```



```

ROUND(

COUNT(CASE WHEN Order_Status = 'Returned' THEN 1 END)

/ COUNT(*) * 100,

2

) AS Return_Rate

FROM clean_lagosmart_data

GROUP BY Product_Category

ORDER BY Return_Rate DESC;

```

Business Purpose:

Identifies product categories had the highest return rates.

Appendix B — Technical Notes

Transaction Note 1 — ID Validation

Transaction IDs were checked for duplicates during data-quality validation. Repeated IDs were investigated at record level to distinguish true duplicate transactions from records sharing an ID but containing different attribute values. Confirmed duplicate records were removed before the final analytical dataset was produced.

Technical Note 2 — Total Orders Calculation

Total Orders was calculated using a distinct count of Transaction ID to avoid double-counting transactions where duplicate or repeated identifiers were present.

Technical Note 3 — Financial Validation

Sales, Cost, Profit and Profit Margin were independently validated during data preparation. Profit Margin was calculated as Profit divided by Sales and rounded appropriately during validation to minimize discrepancies caused by decimal precision.

Technical Note 4 — Customer Rating

Customer Rating analysis uses the average customer rating rather than the sum of ratings, as the objective is to measure the typical customer satisfaction level across categories, regions and customer groups.